A3-PS035-TC490

PS-035: Evidence-Based Operator Network Fault Localisation

NEXORA — Evidence-Sufficient and Counterfactual Network Fault Localisation

Azriel Gershom Raj R • **Rovena M**

Faculty Mentor: Mr. Rishab Haldar, M.Tech., Assistant Professor, Division of Computer Science and Engineering, Karunya Institute of Technology and Sciences

AIORI Hackathon 3

Problem Description

One physical fault can generate multiple correlated alarms, making the original fault difficult to localise without combined topology, log and active-measurement evidence [1][3].

- One failure triggers interface changes, OSPF neighbour loss, route recalculation, higher latency, packet loss and reachability problems; severe failures can flood operators with alerts [3].
- Secondary symptoms appear on neighbouring devices, so the loudest alarm may not mark the fault.
- A single ping or log event cannot isolate the cause; redundant paths can hide a fault.
- Topology, logs and active measurements must be weighed together [1][4].

PS-035 requirements

- Five-node instrumented network, five seeded faults
- Combine topology, logs and active ping
- Rule-based baseline before any optional AI
- Ranked Top-1 / Top-3 faults with evidence
- Held-out evaluation
- Report when evidence is insufficient
- No automated remediation

Proposed solution. NEXORA ranks probable faults, verifies competing hypotheses, probes safely, and reports DIAGNOSED, AMBIGUOUS or INSUFFICIENT_EVIDENCE with an evidence certificate.

Background

Technical context needed to understand the problem and the proposed approach

Operations monitoring: syslog events, interface/routing state and active probes are the main evidence sources; monitoring views can be validated by complementary measurements [1].

Topology dependencies: a link fault touches two endpoints; a router fault touches every attached link.

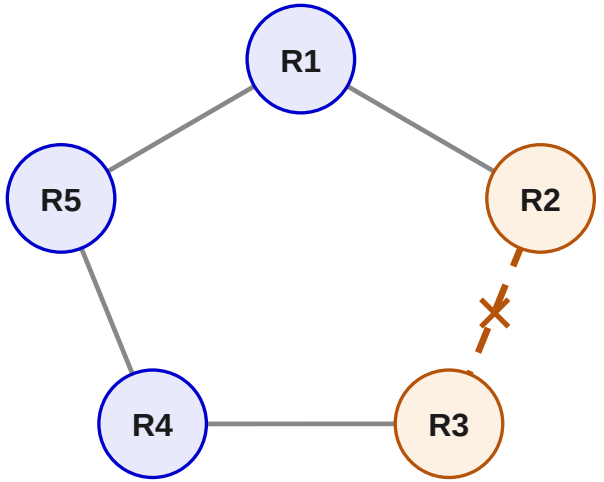
Primary vs secondary: one root fault yields many downstream alarms on other devices; alert floods are documented in large networks [3].

Link vs router failure: both drop adjacencies at neighbours; they differ in what the rest of the router still does; path-aware probing can help separate them [4].

Convergence and failover: OSPF recomputes routes after adjacency loss, so traffic can survive on a redundant path.

Ambiguity: missing, delayed or duplicate observations allow several faults to fit the same evidence; ranked candidates with explanations support operator review [5].

Illustrative example: R2–R3 link failure in the proposed ring



Observed: interface-down logs and OSPF neighbour loss at R2 and R3; R2→R3 traffic now takes R2–R1–R5–R4–R3 with more hops and latency.

Why reachability is not enough: R2 can still ping R3 over the longer path, so a ping-only monitor sees “up” while the primary fault persists.

Literature & RFC Review

Recent research (2023–2026) on monitoring, alarm correlation, probing and fault localisation

Research Paper	Existing Contribution	Relevance to NEXORA
[1] Magnifier (NSDI 2023)	Improves network visibility using complementary monitoring and measurement validation.	Evidence collection and observation verification.
[2] NetAssistant (NSDI 2024)	AI-assisted operator troubleshooting workflows.	Structured diagnosis and operator decision support.
[3] SkyNet (SIGCOMM 2025)	Multi-source alert grouping, severity analysis and alarm-flood interpretation.	Topology-aware event correlation and secondary-alarm reduction.
[4] ByteTracker (SIGCOMM 2025)	Path-aware network probing and failure localisation.	Targeted active measurements and alternative-path reasoning.
[5] BiAn (SIGCOMM 2025)	LLM-assisted network device ranking and diagnostic explanations.	Comparison with our deterministic, evidence-supported ranking approach.
[6] C-ADL (JSS 2026)	Formal architecture-based causal and counterfactual reasoning for distributed systems.	Conceptual foundation for hypothesis verification; it addresses distributed-system architectures, not IP network fault diagnosis.

Research gap. Recent research demonstrates advances in network observability, alarm correlation, active probing, and AI-assisted fault localisation. NEXORA proposes a controlled and reproducible framework integrating deterministic hypothesis ranking, counterfactual symptom verification, targeted read-only probes, and explicit evidence-sufficiency assessment.

Proposed Solution

NEXORA: five-stage evidence-based diagnostic pipeline, read left to right

1 Instrumented Lab & Evidence Collection

- Five FRRouting routers in an OSPF ring (Containerlab)
- Faults injected by interface control and tc/netem
- Collect syslog, interface/OSPF state, topology and active pings [1]

MANDATORY

2 Topology-Aware Event Correlation

- Normalise events to one schema with timestamps
- Group by time window and topology adjacency [3]
- Separate primary from secondary symptoms

MANDATORY

3 Rule-Based Hypothesis Ranking

- Candidates: link, router, OSPF mismatch, loss, delay
- Deterministic Python rules score and rank each candidate [5]
- Top-1 / Top-3 before any ML

MANDATORY

4 Counterfactual Verification & Safe Probing

- Predict each candidate's symptoms on the topology graph; compare with observations [6]
- Tied candidates: run a predefined read-only probe that best separates them [4]

PROPOSED

5 Diagnosis & Uncertainty Reporting

- Status: DIAGNOSED, AMBIGUOUS or INSUFFICIENT_ EVIDENCE
- Certificate: supporting, contradicting and missing evidence, timestamps, rules [2]
- No automated remediation

OUTPUT + PROPOSED

Mandatory implementation | Proposed enhancement | Optional ML comparison (offline Random Forest)

Seeded faults: F1 R2–R3 link failure • F2 R3 router failure/isolation • F3 OSPF hello-interval mismatch at an R4 interface • F4 packet loss on R4–R5 • F5 delay and jitter on R1–R5.

A Random Forest prediction is never treated as physical evidence; ground-truth labels never enter diagnostic inputs.

Optimization Proposed

Before vs after: proposed engineering improvements and their intended benefits

Existing / Basic Approach	Proposed Nexora Enhancement	Intended Benefit
Independent alarm interpretation	**1 Topology-aware event correlation [3]**	Fewer secondary alarms mistaken for root causes
Static diagnosis ranking	**2 Counterfactual hypothesis verification [6]**	Better discrimination between router and link faults
Fixed measurement sequence	**3 Hypothesis-discriminating active probing [4]**	Uncertainty reduced with targeted, bounded measurements
Forced diagnosis	**4 Evidence sufficiency and abstention**	Abstains rather than guessing on weak evidence
Unstructured explanations	**5 Minimal auditable evidence certificate [2][5]**	Traceable, reviewable reasoning for each ranking
Connectivity-only monitoring	**6 Failover-aware diagnosis [4]**	Detects faults even when ping succeeds via an alternate path

NEXORA's contribution is an integrated, reproducible experimental workflow built on established techniques [1]–[6], not techniques invented from scratch. Benefits are intended; no gain in accuracy or latency has been demonstrated.

From Correlation to Evidence-Driven Verification.

Solution Architecture & Design

Proposed layered component architecture (design illustration, not a deployed system)

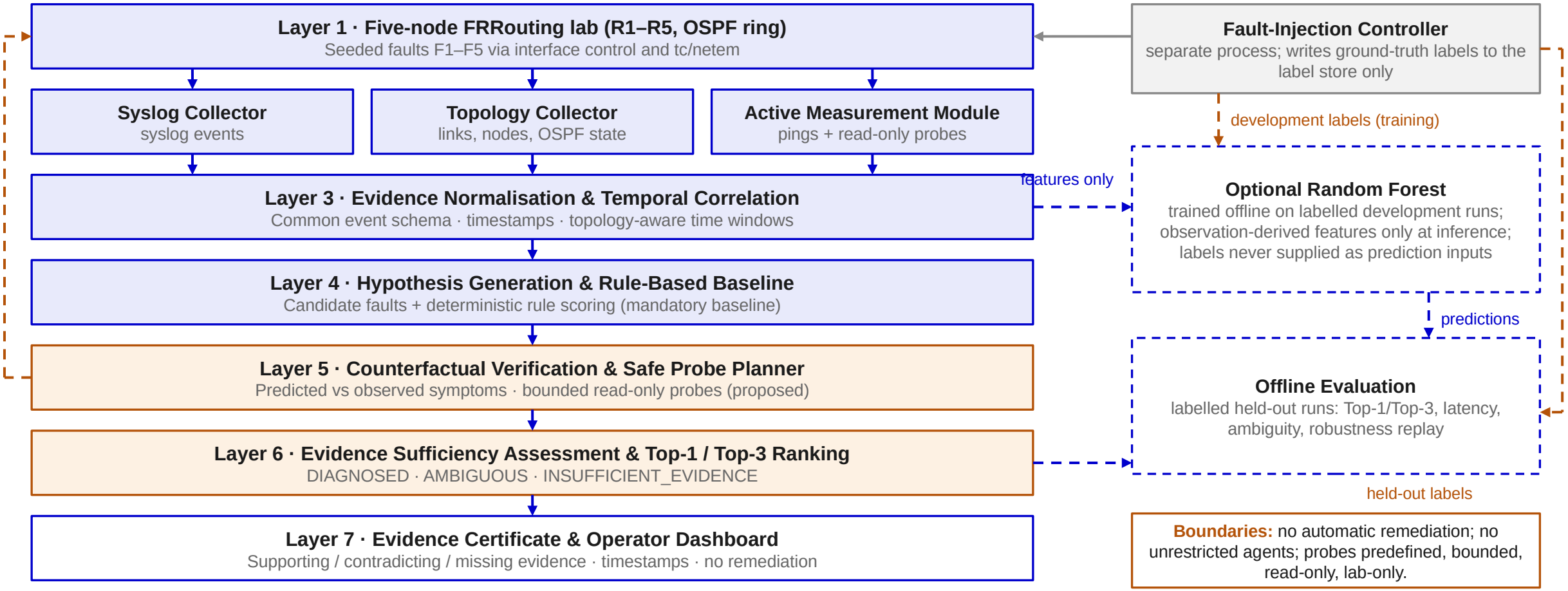


Figure: Proposed architecture. Solid: live evidence flow. Dashed blue: optional/offline. Dashed amber: probe requests (left loop) and ground-truth labels (right): development labels reach RF training only; held-out labels reach offline evaluation only.

Data Flow Diagram

Proposed data movement between processes (P), stores (D) and external entities (E)

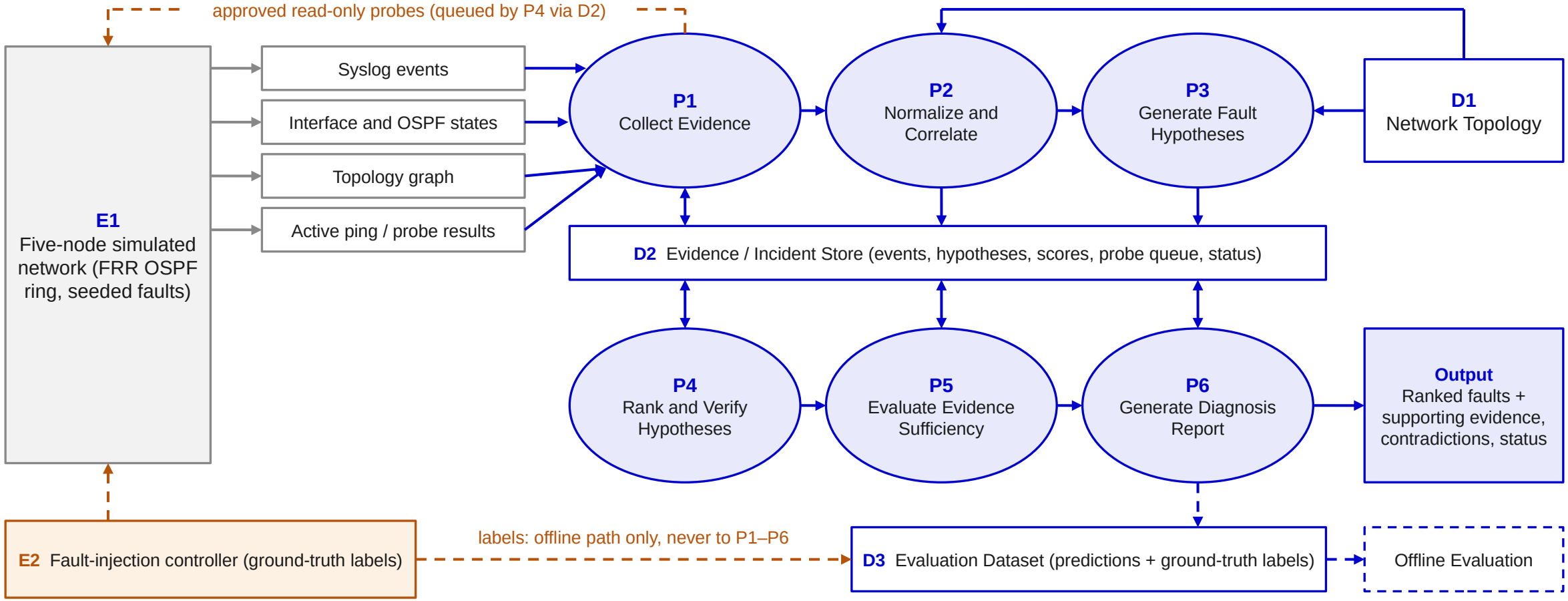


Figure: Data Flow Diagram (proposed). Probe requests are queued in D2 and executed read-only by P1; no flow leads from diagnosis to device configuration; labels reach D3 only.

Deployment Timeline

Phased roadmap: working baseline first, enhancements and optional features after

Phase 0
Repository, specification and topology design

BASELINE

Phase 1
Five-router lab and fault injection

BASELINE

Phase 2
Syslog, topology and active measurement collection

BASELINE

Phase 3
Rule-based diagnosis and Top-3 ranking

BASELINE

Phase 4
Counterfactual verification, active probing and uncertainty handling

ENHANCEMENT

Phase 5
Held-out evaluation and robustness experiments

EVALUATION

Phase 6
Optional ML comparison and operator dashboard

OPTIONAL

Phase 7
Documentation, final validation and demonstration

FINAL

Evaluation plan

- Repeat each of F1–F5, varying fault location and timing
- Separate development and held-out runs
- Top-1 / Top-3 accuracy and diagnostic latency
- Ambiguity and insufficient-evidence cases
- Baseline vs enhanced variants vs optional Random Forest
- Robustness replay: missing, delayed, duplicate, conflicting observations; no ground-truth leakage

Bibliography

[1] T. Bühler, R. Jacob, I. Poese, and L. Vanbever. “Enhancing Global Network Monitoring with Magnifier.” In: *Proc. 20th USENIX Symposium on Networked Systems Design and Implementation (NSDI)* (2023), pp. 1521–1539. https://www.usenix.org/conference/nsdi23/presentation/buhler

[2] H. Wang et al. “NetAssistant: Dialogue Based Network Diagnosis in Data Center Networks.” In: *Proc. 21st USENIX Symposium on Networked Systems Design and Implementation (NSDI)* (2024), pp. 2011–2024. https://www.usenix.org/conference/nsdi24/presentation/wang-haopei

[3] B. Yang et al. “SkyNet: Analyzing Alert Flooding from Severe Network Failures in Large Cloud Infrastructures.” In: *Proc. ACM SIGCOMM 2025 Conference* (2025), pp. 512–526. DOI: 10.1145/3718958.3750536

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[5] C. Wang et al. “Towards LLM-Based Failure Localization in Production-Scale Networks.” In: *Proc. ACM SIGCOMM 2025 Conference* (2025), pp. 496–511. DOI: 10.1145/3718958.3750505

[6] M. Tanhaei. “C-ADL: A Causal Architecture Description Language for Design-Time Root Cause Analysis and Counterfactual Reasoning in Distributed Systems.” In: *Journal of Systems and Software* 239 (2026), Article 112902. DOI: 10.1016/j.jss.2026.112902

Technical standards and implementation documentation (RFC 5424, RFC 8345, FRRouting, Containerlab, tc-netem) are not peer-reviewed papers and are listed in the project README.

THANK YOU

NEXORA

From Correlation to Verification.

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